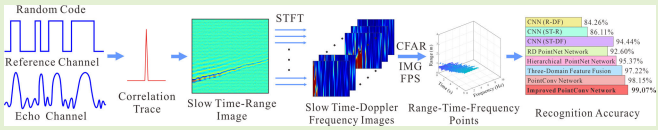
Random Code Radar With Range–Time–Frequency Points and Improved PointConv Network for Through-Wall Human Action Recognition

Hang Xu, Yong Li, Qingran Dong, Li Liu, Jingxia Li, Jianguo Zhang, and Bingjie Wang

***Abstract*—We proposed and demonstrated experimentally a random code radar with range–time–frequency points and the improved PointConv network for through-wall human action recognition (HAR). The physical random code signal with the natural random and aperiodicity is used as the radar-transmitted waveform. A series of slow time-Doppler frequency (ST-DF) images are obtained by correlation ranging and short-time Fourier transform (STFT) for echo and reference signals, and then are arranged at different ranges to obtain the 3-D range–time–frequency matrix. The range–time–frequency points are input into the improved PointConv network after the constant false alarm rate (CFAR) detection, isosurface mesh generation (IMG), and farthest point sampling (FPS) for the 3-D matrix. The PointConv network is improved by model simplification and structural enhancement, which can achieve higher recognition accuracy, the smaller parameters with 5.53 M, and the smaller floating-point operations (FLOPs) of 1.06 G, compared to the existing PointConv network. Experimental results demonstrate that the proposed radar can accurately recognize human actions behind walls with a 99.63% average accuracy for ten actions and a 96.83% average accuracy for six participants. Compared with the 2-D image-based convolutional neural network, three-domain feature fusion, two 3-D point-based PointNet networks, and 3-D point-based PointConv network, the proposed method realizes the higher recognition accuracy.**



***Index Terms*— Human action recognition (HAR), PointConv, random code radar, range–time–frequency points.**

I. Introduction

HUMAN action recognition (HAR) has a wide application prospect in anti-terrorism action, social order maintenance, medical monitoring, and smart homes. Wearable sensors such as the accelerometer, gyroscope, and magnetometer [1], [2], [3] have large-scale applications in HAR. They obtain abundant and valuable action information by detecting the changes in acceleration, angular velocity, and magnetic field caused by the wearer's action. However, they are not suitable for noncooperative human targets and have several deficiencies including being uncomfortable to wear, easily damaged, and forgotten. In addition, the external sensors including the pyroelectric infrared sensor [4], video sensor [5], and radar [6] are also used for HAR. Different from infrared-based and video-based sensors, the radar can operate normally in harsh environmental temperatures, visibility, and through-obstacle scenarios. It transmits various electromagnetic waves as probe signals for HAR, mainly including the single-frequency continuous wave [7], frequency-modulated continuous wave [8], ultrawideband pulse [9], stepped-frequency continuous wave [10], and random code signal [11]. Compared with other types of transmitted signals, the random code signal has a lower range sidelobe and stronger anti-jamming performance [12], which benefits from its good autocorrelation property. Moreover, it also has a wide bandwidth, high range resolution, and good electronic counter countermeasure ability. Therefore, the random code radar has been successfully used in through-wall imaging [13], vital signs detection [14], and buried pipe location [15]. Especially, its advantages have been demonstrated in through-wall HAR [16]. However, the simple feature extraction and fusion methods based on the 2-D radar images do not make full use of human action features captured by the random code radar, thus limiting the further improvement of HAR accuracy.

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In radar-based HAR methods, the 3-D joint-variable representation simultaneously contains the human action information in the range, slow time, and Doppler frequency domains. Compared with the 2-D joint-variable representation such as the time-frequency domain, it more comprehensively and intuitively reveals the potential dependencies and correlations among the three variables of action features. He et al. [17] proposed the range-Doppler surface (RDS) to analyze the human action feature more comprehensively. However, this method lacks the ability to extract targets separately [18]. Du et al. [18] designed the 3-D model with range-velocity-time points to separate the aggregate features of multiple human targets. Erol and Amin [19] proposed the radar data cube (RDC)-based multilinear subspace method to identify fall and nonfall with 97.88% recognition accuracy. They later proposed the boosting scheme for RDC-based processing and obtained the recognition accuracy of 97.2% for five-class human actions [20]. Du et al. [21] further proposed the 3-D deep learning framework for HAR using the range-Doppler (RD)-time points and RD PointNet network. Its average accuracy for HAR in free space reached 87.16% for six-class actions and five participants. However, this approach lacks the ability to obtain the local structures of human micromotion features based on the hierarchical way [22]. To solve this problem, Li et al. [22] proposed the temporal RD PointNet-based approach, which feeds range-velocity-time points into the hierarchical PointNet network. Its average accuracy for HAR behind the opaque wooden board reached 90.35% for six class actions and five participants. However, the RD PointNet network and hierarchical PointNet network aggregate the human action feature by the max-pooling layer, which only extracts the strongest features across the local or global regions, potentially losing valuable detailed information for classification and recognition tasks. Wu et al. [23] proposed the PointConv, which extends the convolutional operation into the point clouds for the intact feature transmission and automatically approximates the model weight by the multilayer perceptron (MLP). In 2023, the PointConv network was successfully applied to the pattern recognition of quartz sand particles, showing the outstanding ability to capture the global and local features of particles [24]. However, it faces the challenge of large parameters (Params) and floating-point operations (FLOPs).

In this article, we propose a random code radar with range–time–frequency points and the improved PointConv network for through-wall HAR. The major contributions of this method can be summarized as follows.

1) Although the advantages of random code radar have been proved in through-wall HAR, the simple feature extraction and fusion methods limit the further improvement of HAR accuracy. Therefore, the range–time–frequency points and improved PointConv network are first introduced into through-wall HAR via the random code radar to improve the recognition accuracy.
2) The PointConv network is improved by model simplification and structural enhancement, which can achieve higher recognition accuracy than the RD PointNet network [21], hierarchical PointNet network [22], and PointConv network [24]. In addition, it has smaller Params and FLOPs compared with the PointConv network.

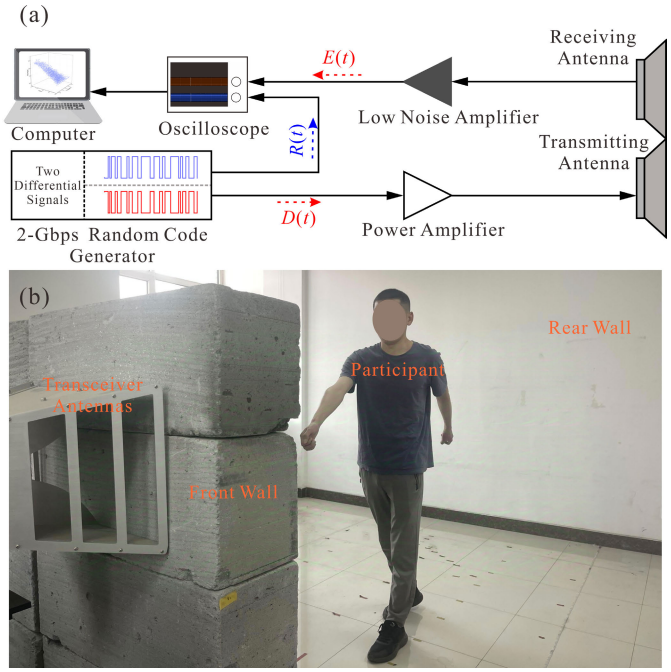

Fig. 1. (a) System structure of random code radar. (b) Detection scenario of through-wall HAR.

II. Dataset Construction

The through-wall HAR dataset is collected by the random code radar. Its system structure is shown in Fig. 1(a). The 2-Gb/s random code generator simultaneously generates two differential signals, namely the detection signal $D(t)$ and the reference signal $R(t)$. The former is transmitted through a wideband horn antenna after being amplified to 14.8 dBm by a power amplifier, while the latter is connected to the oscilloscope for the reference signal acquisition. The echo signal $E(t)$ from the detection scenario is received by another wideband horn antenna, and then it is also sampled by the oscilloscope after being amplified by a low noise amplifier. The oscilloscope continuously collects the signals and transmits them to the computer for data processing and result displaying. The effective bandwidth of the radar system is 1.5 GHz (0.5–2 GHz), which is limited by the operating frequency bands of devices including the oscilloscope with 2 GHz bandwidth and the transceiver antennas with the frequency band of 0.5–3 GHz, and the corresponding range resolution is 10 cm. In the detection scenario shown in Fig. 1(b), the participant performs various actions in the detection space formed by the front wall and rear wall with the 3.9-m spacing. The transceiver antennas are placed facing the participant, separated by the front wall with 20 cm thickness.

Ten human actions identified in this article include [see Fig. 2(a)] squatting, [see Fig. 2(b)] turning, [see Fig. 2(c)] picking up, [see Fig. 2(d)] jumping, [see Fig. 2(e)] boxing, [see Fig. 2(f)] stepping, [see Fig. 2(g)] running forward, [see Fig. 2(h)] walking forward, [see Fig. 2(i)] running backward,

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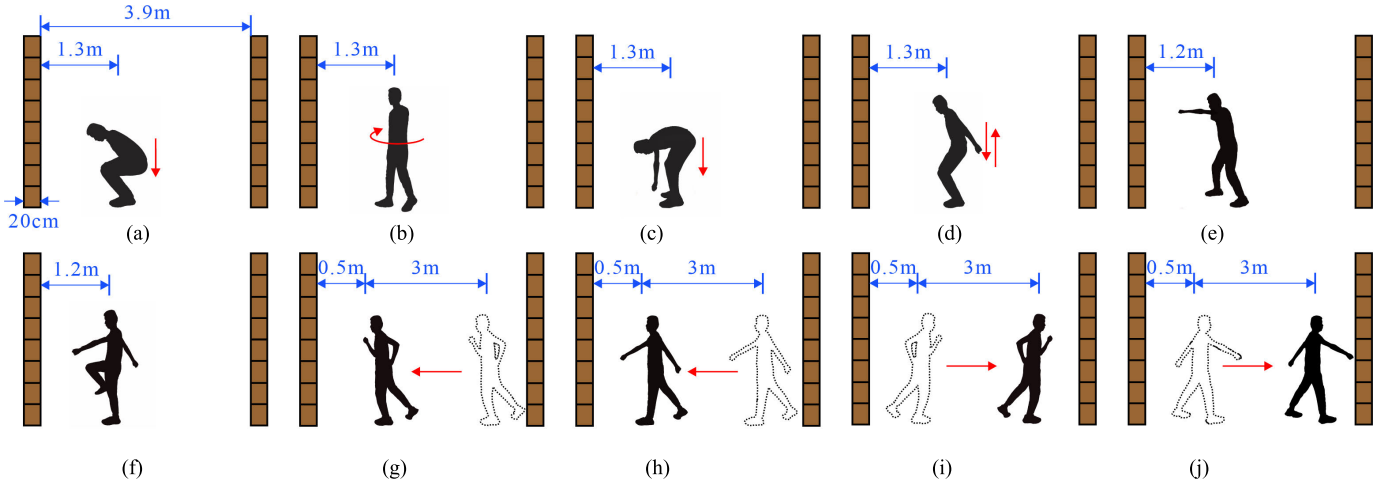

Fig. 2. Diagrams and displacement scopes of ten actions behind walls. (a) Squatting. (b) Turning. (c) Picking up. (d) Jumping. (e) Boxing. (f) Stepping. (g) Running forward. (h) Walking forward. (i) Running backward. (j) Walking backward.

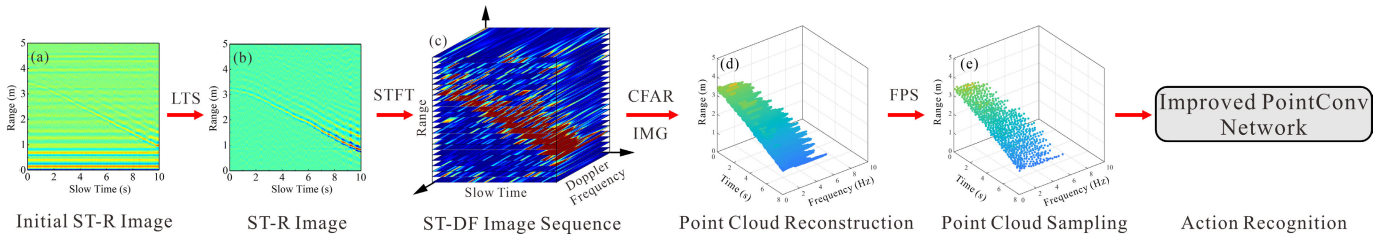

Fig. 3. Through-wall HAR method based on range–time–frequency points and the improved PointConv network. (a) Initial ST-R image. (b) ST-R image. (c) ST-DF image sequence. (d) Point cloud reconstruction. (e) Point cloud sampling.

and [see Fig. 2(j)] walking backward. Fig. 2 shows the diagrams and displacement scopes of ten actions behind walls. A total of 3600 action samples are obtained by six participants, and each participant performs 20 repetitions for each action to get 1200 samples which are then expanded by three times. A more detailed description of action samples is given in [16].

III. Through-Wall HAR Method

In this section, the proposed through-wall HAR method based on range–time–frequency points and the improved PointConv network is described in detail. As illustrated in Fig. 3, the proposed method mainly includes the slow time-Doppler frequency (ST-DF) image sequence, range–time–frequency point reconstruction and sampling, and PointConv-based HAR.

A. ST-DF Image Sequence

The cross correlation calculation is first performed on the signals of reference and echo channels to obtain the correlation trace. The correlation peak is formed by the reflected signal from the participant, so the delay time corresponding to the correlation peak is the round-trip time between the participant and radar. The delay time is calibrated by subtracting the transmission time of the signal inside the radar system and front wall, and then multiplied by the speed of light to obtain the participant's range information behind the front wall. Then, the 2-D slow time-range (ST-R) image in Fig. 3(a) is obtained by the correlation trace accumulation along the slow time. As depicted by Fig. 3(b), to eliminate the direct path wave between the transceiver antennas and the static clutter reflected by walls, the linear trend subtraction (LTS) [25] is applied to the initial ST-R image. Furthermore, by dividing the slow time of 0–10 s into three equal time periods of 0–8, 1–9, and 2–10 s, the number of ST-R images tripled from 1200 to 3600. To extract the Doppler frequency features of human actions, each range unit of the ST-R image is implemented with the short-time Fourier transform (STFT) for obtaining a series of ST-DF images at different ranges that are the ST-DF image sequence, as plotted in Fig. 3(c).

B. Range–Time–Frequency Point Reconstruction and Sampling

These ST-DF images are arranged according to different ranges to obtain the RDC(r, s, d), as shown below

$$\mathrm{RDC}(r, s, d) = \left| \sum_{l=0}^{L-1} \mathbf{TR}(r, s+l) h(l) e^{-j2\pi dl/L} \right|^2 \tag{1}$$

where r, s, and d represent the range domain, slow time domain, and Doppler frequency domain, respectively. l and L represent the index and total number of the sliding window of STFT, respectively. $\mathbf{TR}$ denotes the ST-R matrix, and $h(l)$ is the hamming window function with a window length of 8 and an overlap of 87.5%. The RDC is the 3-D matrix with three coordinate axes namely the range, slow time, and Doppler frequency, so it displays the intact action features in the range, slow time, and Doppler frequency domains. However, the RDC contains large amounts of data and is directly processed using the 3-D convolutional neural network (CNN), which requires

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a large storage space, high hardware performance, and long computational time. Therefore, it is very necessary to reduce the data volume of RDC to retain the valuable information of human action and remove the redundant information.

In the preliminary processing, the strong clutter including the direct path wave and static clutter have been removed by the LTS. Therefore, in the RDC, the signal intensity reflected by the human target is dominant, and its corresponding values are significantly higher than that of noises, clutters, and harmonics between the human target and radar. The point cloud data is arranged in order of intensity from large to small, the top 20% of high-intensity data is retained as the valuable action information, and the remaining data is set to zero as the redundant information. Then, the 2-D constant false alarm rate (CFAR) detector [26] is slid across each ST-DF image pixel by pixel to obtain the pixels with intensities exceeding the threshold. It is used as the sliding window composed of the test pixel, guard window, and clutter window from inside to outside. The test pixel is located at the center of the sliding window, the clutter window estimates the intensity of background clutter to offer the threshold, and the guard window is designed to be the barrier that separates the test pixel and clutter window. After 2-D CFAR detection, the detected scatters of each ST-DF image are regarded as point clouds in the range–time–frequency space. Furthermore, the isosurface mesh generation (IMG) is implemented by creating a surface that has the same intensity value within the ST-DF image sequence. The point cloud reconstruction in Fig. 3(d) is completed. After the above processing, the point clouds of ten human actions have different sizes and uneven distributions. In particular, the point cloud distribution at the edge of the feature contour is relatively sparse. To solve the above problems, the point cloud is sampled to 1024 sampled points using the farthest point sampling (FPS), as shown in Fig. 3(e). Its core idea is to repeatedly place the next sampled point in the center of the least-known area of the sampling manifolds. The process of FPS not only aggregates the valuable information of human action, but also further reduces the data processing volume. The specific implementation steps of FPS are as follows.

1) Randomly select a point p_0 as the starting point from the point clouds with P points generated by the process of IMG to obtain the set of sampled points $S = \{p_0\}$.
2) Calculate the distances from the remaining $P - 1$ points to p_0, and deposit them to array L. Then, select the point p_1 with the maximum distance from L, and add p_1 to S for updating $S = \{p_0, p_1\}$.
3) Calculate the distances from the remaining P–i points to each point in S, and add each minimum distance to the array L. Then, select the point p_i with the maximum distance from L, and add p_i to S to update $S = \{p_0, p_1, \ldots, p_i\}$.
4) Repeat step (3) until 1024 sampled points are generated.

C. PointConv-Based HAR

The improved PointConv network is proposed to classify and recognize the point clouds of ten human actions behind walls, which is optimized and improved based on the PointConv network [24] as follows.

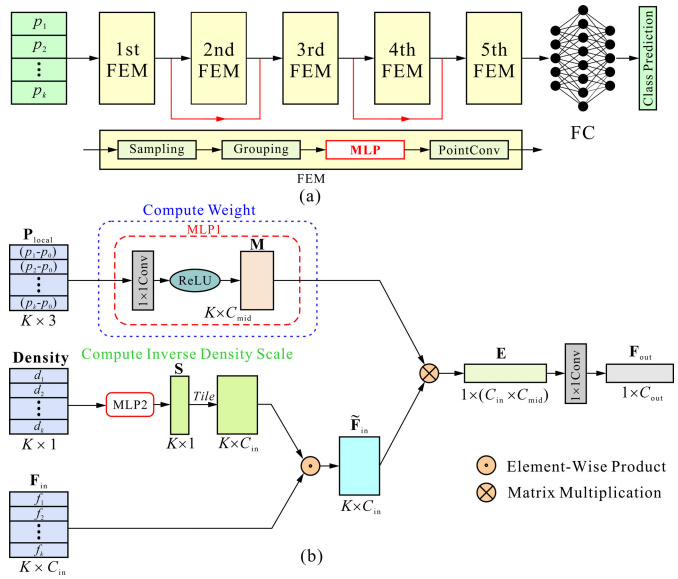

Fig. 4. (a) Training process of the improved PointConv network. (b) Architecture of the PointConv layer.

1) The number of feature encoding modules (FEMs) is increased from 3 to 5, which makes the learning of action features more in-depth and thus improves the HAR accuracy, but also causes the increase of model's Params and FLOPs.
2) The MLP for feature extraction and conversion is reduced from three layers to two layers, omitting a hidden layer, and its final output feature dimension is reduced from 1024 to 512, which simplifies the model and thus reduces the Params and FLOPs, but also causes the reduction of HAR accuracy.
3) The residual connections are added to enhance the network structure and the learning ability of human action features. Finally, based on the joint optimization of the above three improvement strategies, the Params and FLOPs of the network are reduced; meanwhile, the high HAR accuracy is maintained, thus achieving the best balance between the network performance and computational complexity.

The training process of the improved PointConv network is shown in Fig. 4(a), it aggregates the action features across the whole point clouds by the hierarchical structure, which contains five FEMs and a fully connected (FC) layer. Each FEM includes a sampling layer, a grouping layer, and a PointConv layer. The sampling layer uses the FPS to sample the action point set, and then the grouping layer groups and abstracts the action point set to construct multiple local regions. Finally, the action features of these local regions are captured by the PointConv layer to obtain the smaller action point subset with the higher dimensions. It should be noted that the MLP for feature extraction and conversion, which is located between the grouping layer and the PointConv layer, maps the input feature to the higher dimensional feature space by layer-by-layer conversion, thereby increasing the feature

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dimension of the local region. In the first FEM, 512 points with three feature dimensions are sampled via the FPS from every action point set with 1024 points with three feature dimensions, and they are used as query points to build local regions. Through calculating the Euclidean distance between the query point and its neighboring points, $K = 16$ nearest neighboring points are found around the query point, which constitutes the local region with the corresponding query point. The feature dimension is increased from 3 to 64 by the MLP. The feature of each local region is captured by the PointConv layer, and then the feature matrix composed of 512 sampled points and 64 feature dimensions is obtained for each batch and is input into the next FEM. In the second FEM, the sampling and grouping processes are implemented on the 512 sampled points, the query scope is expanded to $K = 24$ nearest neighboring points, the dimensions of MLP are set to (64, 64), and then the feature matrix with 512 sampled points and 64 feature dimensions is obtained for each batch after the PointConv layer. The outputs from the first FEM and second FEM are fused by the first residual connection to obtain the feature matrix containing 512 sampled points and 128 feature dimensions as the feature input for the third FEM. In the third FEM, 256 points are sampled from 512 points and each sampled point is used as the query point to search $K = 32$ nearest neighboring points to construct the larger local region, and then the feature dimension is 128 after the MLP. Through the PointConv layer, the feature dimension of 256 sampled points is 128. In the fourth FEM, the sampling and grouping processes are implemented on the 256 sampled points, the query scope is expanded to $K = 40$ nearest neighboring points, the MLP's dimensions are (128, 128), and then the feature matrix with 256 sampled points and 128 feature dimensions is generated for each batch through the PointConv layer. The second residual connection is used to combine the outputs of third FEM and fourth FEM, and then the feature matrix with 256 sampled points and 256 feature dimensions is obtained and input into the fifth FEM. In the fifth FEM, the query scope covers all input points, and a point feature vector with 512 feature dimensions is generated and input into the FC layer with the dimensions of (512, 512, 256) for class prediction.

As shown in Fig. 4(b), the PointConv layer mainly includes the weight function $W(\delta_x, \delta_y, \delta_z)$ computation, inverse density scale $S(\delta_x, \delta_y, \delta_z)$ computation, and local region feature matrix. In W computation, the W of each local region is automatically approximated by the MLP1 with the 3-D coordinates $(\delta_x, \delta_y, \delta_z)$. The MLP1 is implemented by the 1×1 convolution with the dimensions of (8, 8, 16). To maintain the permutation invariance, the same W of MLP1 is shared across each local region. In S computation, the density of each point is first computed by the kernelized density estimation (KDE). Assuming that the probability density function f has the observations with X_j $(j = 1, \ldots, m)$, so its kernel estimator $f_h(x)$ is given by the following equation:

$$f_h(x) = m^{-1} \sum_{j=1}^{m} K_h\big(x - X_j\big) \tag{2}$$

where $K_h(u) = K(u/h)/h$, h denotes the bandwidth and K denotes the kernel. Then, the density of each point is fed into the MLP2 to implement the nonlinear transformation. The MLP2 is implemented by 1×1 convolution with the dimensions of (16, 8, 1). Finally, the output feature matrix $\mathbf{F}_{\text{out}}$ of the PointConv layer is expressed below

$$\begin{aligned}
\mathbf{F}_{\text{out}} &= \sum_{k=0}^{K-1} \sum_{c_{\text{in}}=0}^{C_{\text{in}}-1} \big[\tilde{\mathbf{F}}_{\text{in}}(k, c_{\text{in}})\mathbf{W}(k, c_{\text{in}})\big] \\
&= \sum_{k=0}^{K-1} \sum_{c_{\text{in}}=0}^{C_{\text{in}}-1} \left\{ \tilde{\mathbf{F}}_{\text{in}}(k, c_{\text{in}}) \sum_{c_{\text{mid}}=0}^{C_{\text{mid}}-1} [\mathbf{M}(k, c_{\text{mid}})\mathbf{H}(c_{\text{mid}}, c_{\text{in}})] \right\} \\
&= \sum_{c_{\text{in}}=0}^{C_{\text{in}}-1} \sum_{c_{\text{mid}}=0}^{C_{\text{mid}}-1} \left\{ \mathbf{H}(c_{\text{mid}}, c_{\text{in}}) \sum_{k=0}^{K-1} \big[\tilde{\mathbf{F}}_{\text{in}}(k, c_{\text{in}})\mathbf{M}(k, c_{\text{mid}})\big] \right\} \\
&= \text{Conv}_{1\times1}\big(\mathbf{H}, \tilde{\mathbf{F}}_{\text{in}}^{\text{T}}\mathbf{M}\big)
\end{aligned} \tag{3}$$

where k is the index for kth neighbor, K is the number of points in each local region, $\tilde{\mathbf{F}}_{\text{in}} = \mathbf{S} \cdot \mathbf{F}_{\text{in}} \in \mathbb{R}^{K \times C_{\text{in}}}$, $\mathbf{S}$ is the inverse density scale matrix, $\mathbf{F}_{\text{in}}$ is the input feature matrix, $\mathbf{H}(c_{\text{mid}}, c_{\text{in}}) \in \mathbb{R}^{C_{\text{out}}}$ is a vector from $\mathbf{H}$, $\mathbf{H} \in \mathbb{R}^{C_{\text{mid}} \times (C_{\text{in}} \times C_{\text{out}})}$ is the weights of the last layer in the same MLP1, $\mathbf{M} \in \mathbb{R}^{K \times C_{\text{mid}}}$ is the input to the last layer in the MLP1 for computing W, and $\text{Conv}_{1\times1}$ is 1×1 convolution. C_{in}, C_{mid}, and C_{out} are the number of channels for the input, middle, and output features, respectively. c_{in} and c_{mid} are the indices of the input and middle layers, respectively.

IV. Experimental Proof

A. Action Data Analysis

Fig. 5 shows the 3-D point cloud distribution of ten human actions behind walls. The colorbar represents the participant's range information behind the front wall. For instance, the participant squats at 1.3 m behind the front wall within the time scope of 4–6 s, accompanied by the slight range shift caused by his torso leaning forward. As shown in Fig. 5(a), along the range axis, the point clouds of squatting are mainly distributed around 1.3 m. Meanwhile, the point clouds with significant changes in the Doppler frequency axis appear in the time scope of 4–6 s, corresponding to the range and time at which this action occurs. It should be explained that after the participant squats, the rear wall at 3.9 m is suddenly exposed to the detection direction of radar and reflects the detection signal to form the echoes, which belong to the nonstatic clutter for the entire observation time and cannot be removed by the LTS. A similar situation occurs when the participant bends down to pick something up, as plotted in Fig. 5(c). Turning and jumping occur at a similar range and time scopes with squatting, both of which belong to the in situ action with the short duration, and their point clouds are shown in Fig. 5(b) and (d). Boxing and stepping belong to the in situ action located 1.2 m behind the front wall and occur repeatedly during the observation time. Therefore, as depicted by Fig. 5(e) and (f), their point clouds are distributed along the entire slow time axis. The point clouds of stepping are mainly distributed around 1.2 m along the range direction, while the distribution space of boxing is larger, due to its larger amplitude of arm swing facing the radar than that of stepping.

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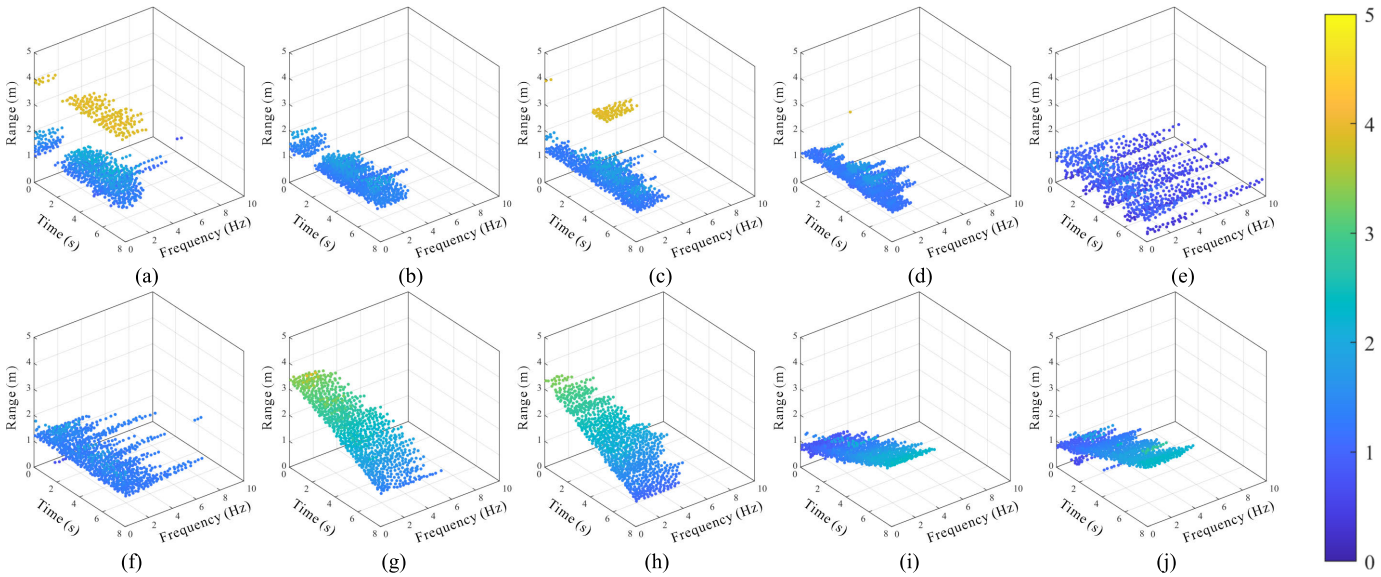

Fig. 5. Range–time–frequency points of ten human actions behind walls. (a) Squatting. (b) Turning. (c) Picking up. (d) Jumping. (e) Boxing. (f) Stepping. (g) Running forward. (h) Walking forward. (i) Running backward. (j) Walking backward.

Moreover, the repeated arm swings also cause their point clouds to exhibit periodic Doppler frequency changes. Similar Doppler frequency changes also occur in swinging arms of walking and running. The frequency of arm swing caused by running is higher than that caused by walking, resulting in the higher occurrence frequency of Doppler frequency peaks for running. As shown in Fig. 5(g) and (h), the point clouds of running forward and walking forward are distributed along the range direction from far to near over time. Fig. 5(i) and (j) shows that the point clouds of running backward and walking backward are distributed along the range direction from near to far over time. During the aforementioned actions except for squatting and picking up, the participant is always located within the 3-dB beamwidth of transceiver antennas, resulting in the detection signal being mainly reflected by the participant's body, so the reflected signal from the rear wall at 3.9 m is very weak, and the weak static clutter caused by the rear wall is removed by the LTS, so the point cloud at 3.9 m is almost absent.

B. Action Recognition Results

The improved PointConv network is implemented on the computer with 32 GB RAM, equipped with an 8 GB GPU and an Intel i9 14-core processor. The 3-D range–time–frequency points are used as the action samples. The number of action samples reaches 3600, which are randomly disrupted and grouped into two parts including 2520 training action samples and 1080 testing action samples, according to the ratio of 7:3. The threefold cross-validation is adopted. The improved PointConv network is trained in a supervised manner, using the stochastic gradient descent optimizer with an initial learning rate of 0.01 and a momentum of 0.9. The batch size is set to 32, and the number of training epochs reaches 100. In the process of network training, the data augmentation strategy is adopted by randomly rotating the point cloud along the range axis and jittering each point by the Gaussian noise with zero mean and 0.01 standard deviation, thereby increasing the data diversity to prevent overfitting.

To demonstrate the effect of improvement strategies in the improved PointConv network, the performances of PointConv networks with different structures are compared in Table I. These networks include the PointConv network in [24] composed of three FEMs, three-layer MLPs, and the final output feature dimension of 1024, the PointConv network 1 composed of five FEMs, three-layer MLPs, and the output dimension of 1024, the PointConv network 2 constituted of five FEMs, two-layer MLPs, and the output dimension of 1024. The PointConv network 3 consists of five FEMs, two-layer MLPs, and the output dimension of 512. The performance of the improved PointConv network is also listed in Table I, which consists of five FEMs, two-layer MLPs, an output dimension of 512, and two residual connections. By comparing the PointConv network in [24] and the PointConv network 1, it can be found that as the number of FEMs increases from 3 to 5, the recognition accuracy of ten human actions behind walls increases to 99.72% due to the more in-depth action feature learning, but it also increases the network complexity, and thus the model's Params and FLOPs increase to 19.96 M and 2.14 G, respectively. The PointConv network 2 with five FEMs reduces the MLPs from three layers to two layers, resulting in smaller Params and FLOPs, but also accompanied by a small decline in the recognition accuracy. The PointConv network 3 reduces the final output feature dimension from 1024 to 512 on the basis of PointConv network 2. Although its recognition accuracy is still further decreased, the Params and FLOPs also show different degrees of decline. The above results show that it is difficult to achieve the balance between the high recognition accuracy and the small Params and FLOPs for the PointConv network. To solve this problem, the improved PointConv network introduces the residual connections on the basis of increasing the FEMs, reducing the MLP layers and feature dimensions, thus, the computational path is simplified and the

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT POINTCONV NETWORKS

PointConv Networks	Number of FEMs	Dimensions of MLPs	Residual Connections	Accuracy (%)	Params (M)	FLOPs (G)
PointConv Network [24]	3	(64, 64, 128), (128, 128, 256), (256, 512, 1024)	No	99.26	19.56	1.17
PointConv Network 1	5	(64, 64, 64), (64, 64, 128), (128, 128, 128), (128, 128, 256), (256, 512, 1024)	No	99.72	19.96	2.14
PointConv Network 2	5	(64, 64), (64, 128), (128, 128), (128, 256), (256, 1024)	No	99.63	19.52	1.64
PointConv Network 3	5	(64, 64), (64, 128), (128, 128), (128, 256), (256, 512)	No	99.44	6.54	1.59
Improved PointConv Network	**5**	**(64, 64), (64, 64), (128, 128), (128, 128), (256, 512)**	**Yes**	**99.63**	**5.53**	**1.06**

TABLE II
HAR ACCURACIES FOR TEN HUMAN ACTIONS USING CNN, FEATURE FUSION, POINTNET, POINTCONV, AND IMPROVED POINTCONV NETWORKS

Methods	Action (%)										Mean
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	
CNN (R-DF) [27]	89.81	99.07	93.52	97.22	99.07	98.15	85.19	77.78	84.26	79.63	90.37
CNN (ST-R) [27]	96.30	89.81	98.15	99.07	100	100	88.89	87.04	86.11	81.84	92.72
CNN (ST-DF) [27]	88.89	93.52	92.60	91.67	97.22	95.37	96.30	96.30	94.44	97.22	94.35
RD PointNet Network [21]	98.15	100	97.22	95.37	97.22	95.37	94.44	88.89	92.60	92.60	95.19
Hierarchical PointNet Network [22]	100	97.22	99.07	98.15	98.15	97.22	97.22	92.60	95.37	90.74	96.57
Three-Domain Feature Fusion [16]	98.15	97.22	99.07	98.14	99.07	100	93.52	94.44	97.22	95.37	97.22
PointConv Network [24]	100	99.07	100	100	100	100	98.15	98.15	98.15	99.07	99.26
Improved PointConv Network	**100**	**100**	**100**	**100**	**100**	**100**	**99.07**	**99.07**	**99.07**	**99.07**	**99.63**

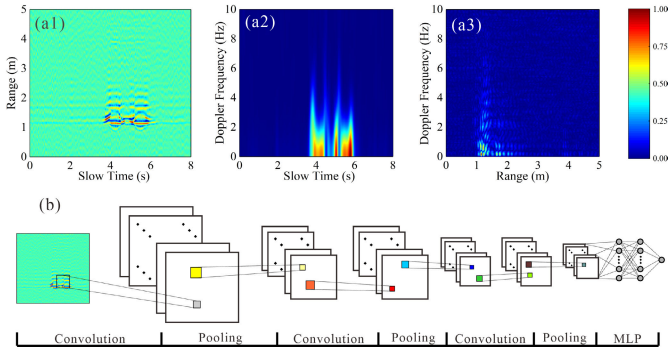

Fig. 6. (a1) ST-R image. (a2) ST-DF image. (a3) R-DF image of jumping. (b) 2-D image-based CNN.

learning ability of action features is enhanced. Therefore, the recognition accuracy, Params, and FLOPs of the improved PointConv network reach 99.63%, 5.53 M, and 1.06 G, and its performance is significantly better than that of the existing PointConv network.

Table II compares the recognition accuracies of ten human actions behind walls based on the different methods, including the 2-D image-based CNN [27], two 3-D point-based PointNet networks [21], [22], three-domain feature fusion [16], 3-D point-based PointConv network [24], and 3-D point-based improved PointConv network. The 2-D radar images input to the CNN include the ST-R image, ST-DF image, and range-Doppler frequency (R-DF) image. Taking jumping as an example, its ST-R image, ST-DF image, and R-DF image are shown in Fig. 6(a1)–(a3), which show its change situations of range with slow time, Doppler frequency with slow time, and Doppler frequency with range. Fig. 6(b) shows the 2-D image-based CNN, and its structure is similar to that in [27]. The CNN consists of three convolutional layers and two FC layers. Each convolutional layer contains 20 convolutional kernels with a size of 5×5. Two FC layers contain 500 hidden units in the hidden layer. The results in Table II show that the average recognition accuracies of CNN based on the R-DF image, ST-R image, and ST-DF image are 90.37%, 92.72%, and 94.35%, respectively. The average recognition accuracy of the three-domain feature fusion method reaches 97.22%. The average recognition accuracies of the RD PointNet network, hierarchical PointNet network, and PointConv network utilizing the 3-D range–time–frequency points are 95.19%, 96.57%, and 99.26%, respectively. The proposed improved PointConv network with 3-D points has the highest average recognition accuracy, reaching 99.63%. The above results indicate that compared to the 2-D radar images, the 3-D range–time–frequency points and three-domain feature fusion can more fully display the human action features in the range domain, slow time domain, and Doppler frequency domain, which is conducive to the accuracy improvement. More importantly, compared with two PointNet networks and the PointConv network, the improved PointConv network can realize higher 3-D recognition accuracy.

Table III shows the confusion matrix of through-wall HAR using the improved PointConv network. Only 0.93% of (g) running forward and (h) walking forward are misjudged as each other. This is because although their Doppler frequency changes caused by different frequencies of arm swing are different, they both have continuous and periodic action features. Similar results also occur between (i) running backward and (j) walking backward.

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TABLE III
CONFUSION MATRIX OF TEN HUMAN ACTIONS USING IMPROVED POINTCONV NETWORK

Pred/True (%)	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
(a)	**100**	0	0	0	0	0	0	0	0	0
(b)	0	**100**	0	0	0	0	0	0	0	0
(c)	0	0	**100**	0	0	0	0	0	0	0
(d)	0	0	0	**100**	0	0	0	0	0	0
(e)	0	0	0	0	**100**	0	0	0	0	0
(f)	0	0	0	0	0	**100**	0	0	0	0
(g)	0	0	0	0	0	0	**99.07**	0.93	0	0
(h)	0	0	0	0	0	0	0.93	**99.07**	0	0
(i)	0	0	0	0	0	0	0	0	**99.07**	0.93
(j)	0	0	0	0	0	0	0	0	0.93	**99.07**

TABLE IV
HAR ACCURACIES FOR SIX PARTICIPANTS USING CNN, FEATURE FUSION, POINTNET, POINTCONV, AND IMPROVED POINTCONV NETWORKS

Methods	Participant (%)						Mean
	P1	P2	P3	P4	P5	P6	
CNN (R-DF) [27]	84.17	87.67	83.50	83.83	84.00	82.67	84.31
CNN (ST-R) [27]	88.50	90.67	86.83	87.33	87.83	87.17	88.06
CNN (ST-DF) [27]	92.17	93.50	89.53	90.67	91.33	89.83	91.17
RD PointNet Network [21]	92.67	95.33	91.67	92.50	92.17	91.83	92.70
Hierarchical PointNet Network [22]	95.17	96.67	93.33	94.67	95.17	92.67	94.61
Three-Domain Feature Fusion [16]	91.17	96.50	92.33	92.33	93.84	92.33	93.08
PointConv Network [24]	95.83	97.67	96.00	96.33	97.00	96.50	96.56
Improved PointConv Network	**96.00**	**97.67**	**96.83**	**96.50**	**97.33**	**96.67**	**96.83**

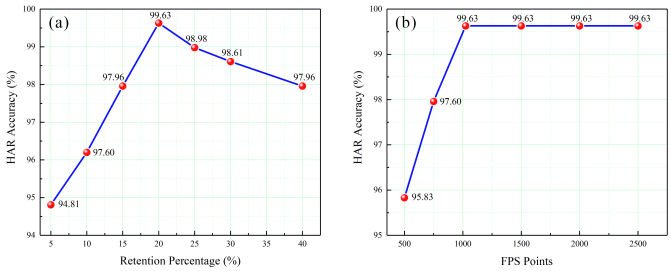

Fig. 7. HAR accuracies with different (a) retention percentages of high-intensity data and (b) FPS points.

The action recognition accuracies for six participants as research targets are listed in Table IV. The point cloud data of any participant are utilized as testing samples, and that of the other five participants are used as training samples. The results show that the average recognition accuracy of six participants using the 3-D point-based improved PointConv network reaches 96.83%, which is higher than that using the 2-D image-based CNN, two 3-D point-based PointNet networks, three-domain feature fusion, and 3-D point-based PointConv network. The comparison results in Table IV ultimately demonstrate that the improved PointConv network with the range–time–frequency points can effectively improve the recognition accuracy of through-wall human action.

Furthermore, the retention percentage of high-intensity data is discussed. Fig. 7(a) shows the influence of different retention percentages on the HAR accuracy. With the increase in retention percentage, the HAR accuracy increases first and then decreases. When the retention percentage is set to 20%, the HAR accuracy reaches the maximum value of 99.63%. This indicates that the top 20% of high-intensity data already contains sufficient and valuable action information, and the retention proportion is further increased to bring the interferences into the point clouds, thereby reducing the HAR accuracy. Therefore, as mentioned earlier, the top 20% of high-intensity data is retained as valuable action information, and the remaining data is set to zero as redundant information. The influence of the number of sampled points in FPS (which is shortened to FPS points) on the HAR accuracy is also discussed, and the number of sampled points is set to 500, 750, 1024, 1500, 2000, and 2500, respectively. The HAR accuracies with the different number of sampled points are given in Fig. 7(b). As the number of sampled points increases, the recognition accuracy also gradually increases. When the number of sampled points increases to 1024, the recognition accuracy stabilizes at 99.63%, indicating that the action information contained in 1024 sampled points is enough to accurately recognize ten actions behind walls without adding additional computational burden. Therefore, as mentioned above, 1024 sampled points are selected for HAR in this article.

V. CONCLUSION

In conclusion, we propose and experimentally prove a random code radar with range–time–frequency points and the improved PointConv network. The random code signal is used as the detection signal for human actions behind walls, and the method is composed of the ST-DF image sequence, range–time–frequency point reconstruction and sampling, and

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improved PointConv network is utilized for through-wall HAR. Compared with the existing HAR methods, the unique advantages of the proposed method include: 1) the 3-D range–time–frequency points display the human action features more comprehensively and efficiently, compared with the 2-D radar images and 2) using the improved PointConv network with 3-D points, the through-wall HAR accuracies reach 99.63% for ten actions and 96.83% for six participants, which are higher than that of 2-D image-based CNN, 3-D point-based PointNet networks, three-domain feature fusion, and 3-D point-based PointConv network. Above all, the improved PointConv network has smaller Params and FLOPs than the existing PointConv network. Therefore, the proposed radar provides a promising solution for HAR behind walls in the application fields of anti-terrorism action, social order maintenance, medical monitoring, and smart homes.

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